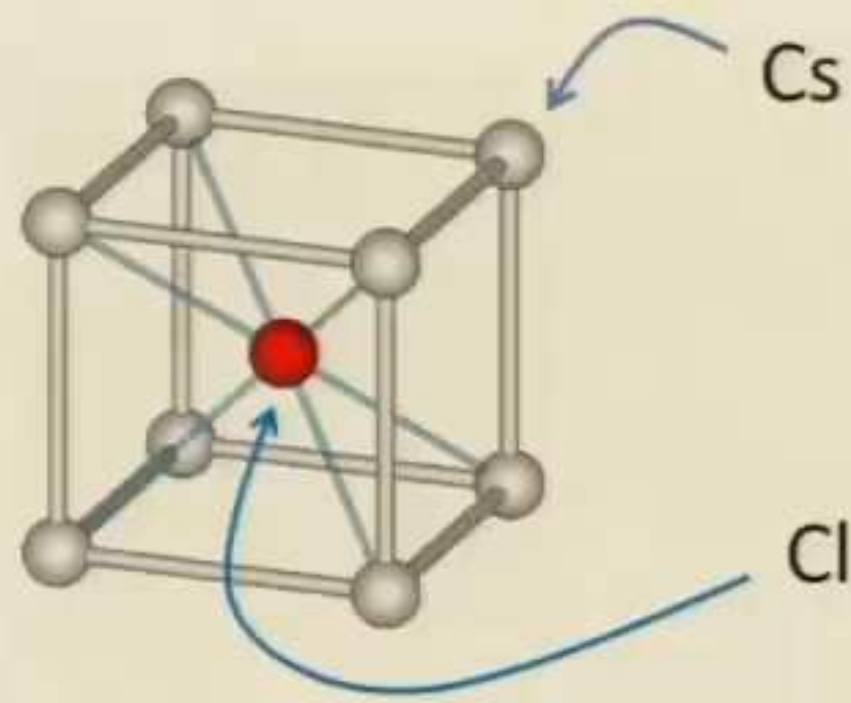




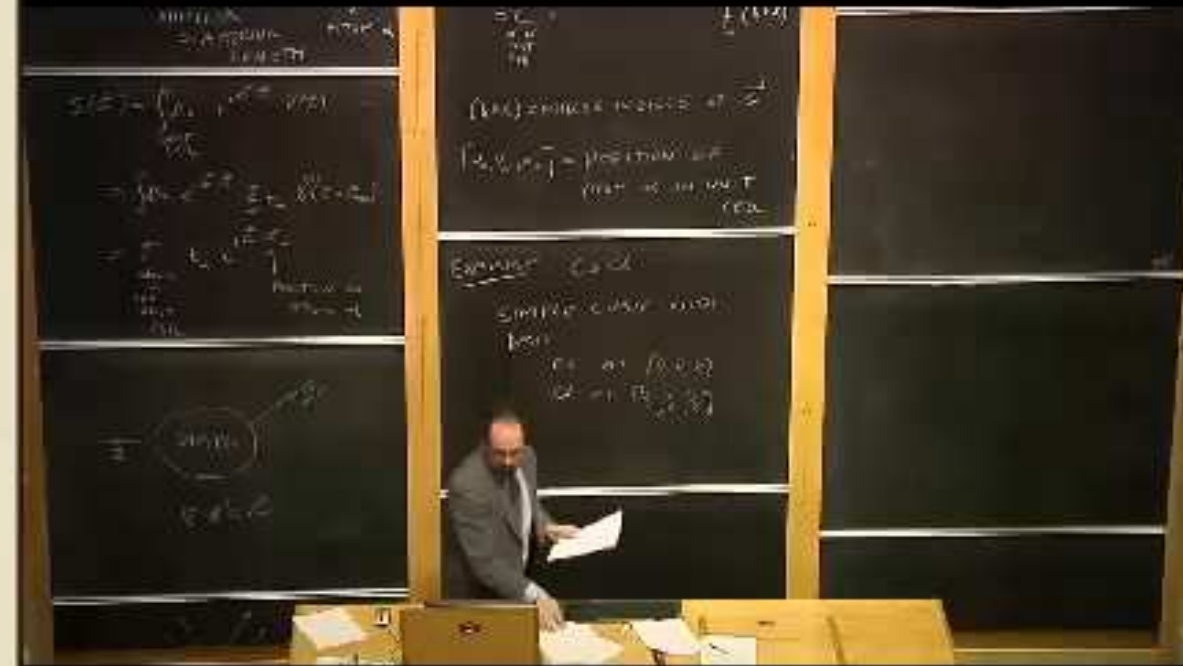


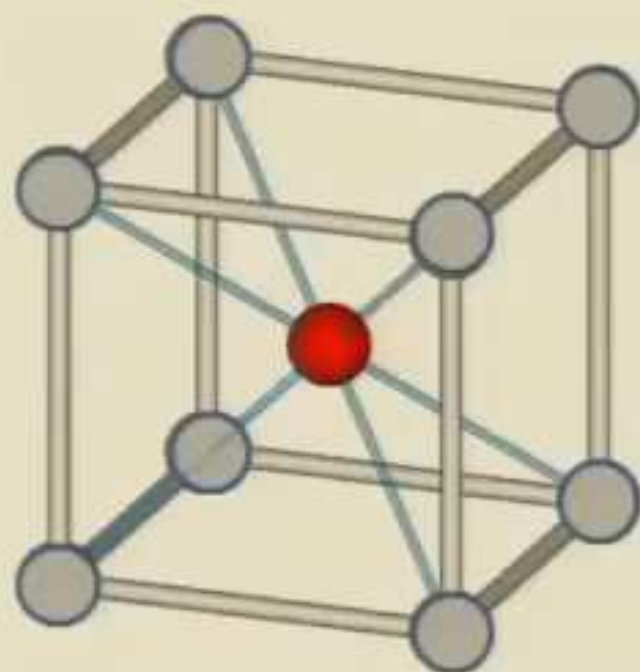
Cesium Chloride (CsCl): A simple cubic Lattice with a Basis

Basis:

Cs at  $[0, 0, 0]$  (i.e., on the simple cubic)

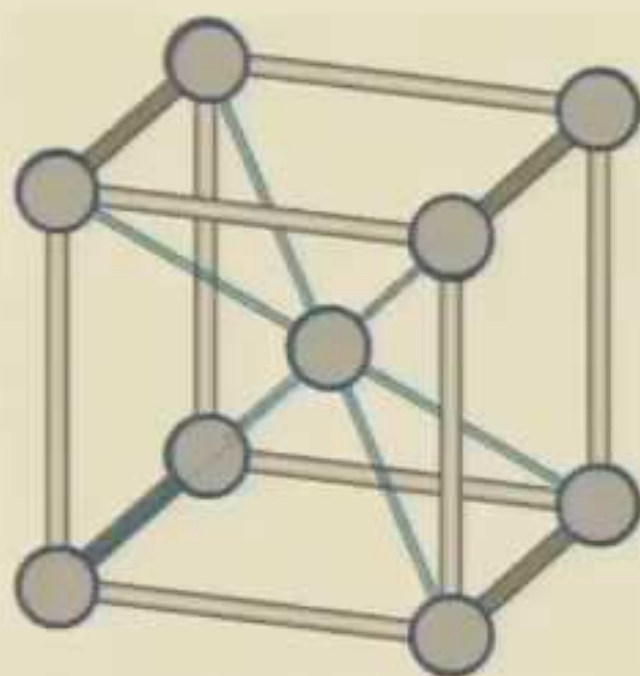
Cl at  $[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$  (i.e., in the middle of each cube)





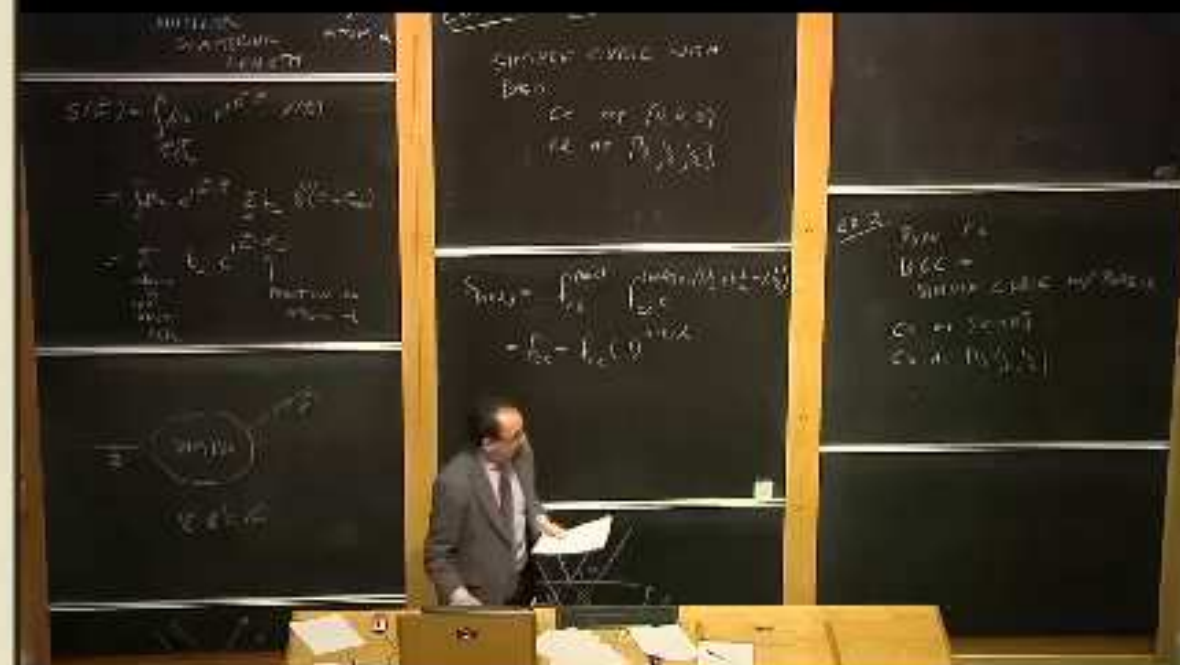
CsCl = Simple Cubic  
with Basis

Cs at  $[0, 0, 0]$   
Cl at  $[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$



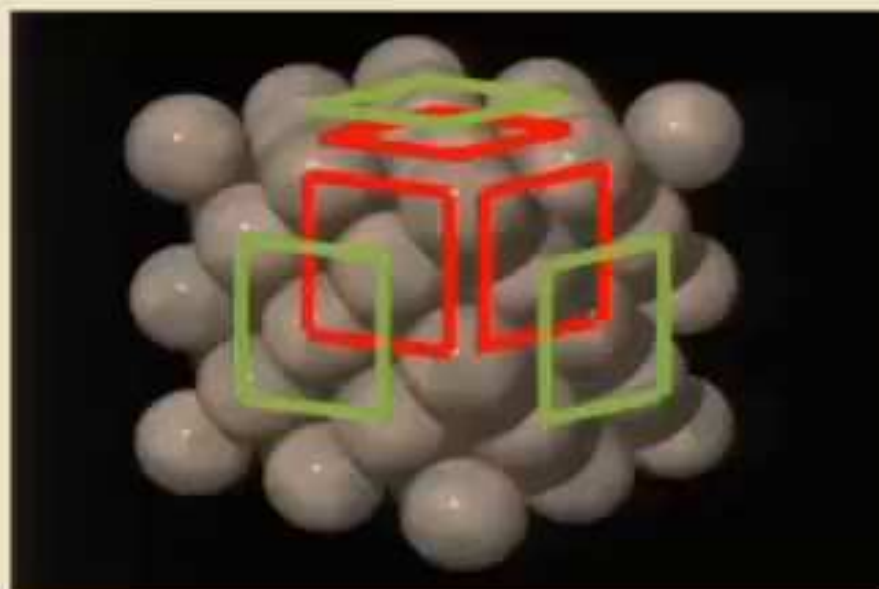
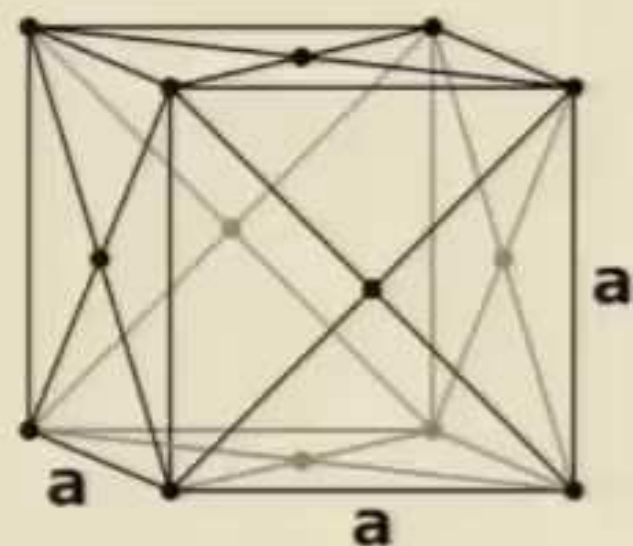
Cs = Simple Cubic  
with Basis  
= BCC

Cs at  $[0, 0, 0]$   
Cs at  $[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$



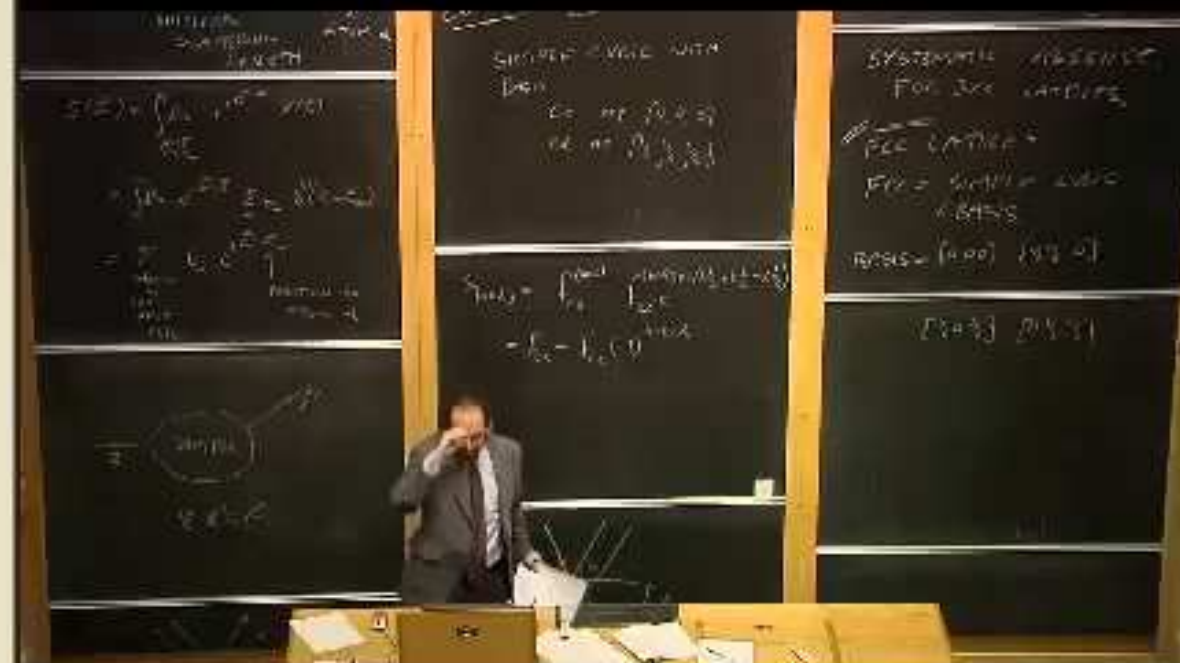


## Unit cell of Face Centered Cubic Lattice (FCC) (Notated cubic-F)

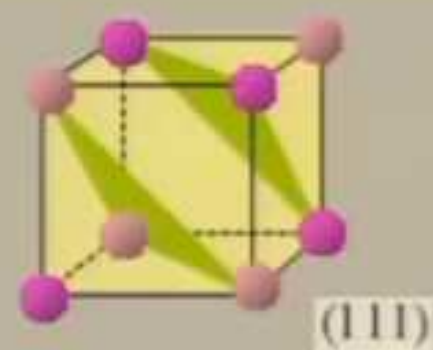
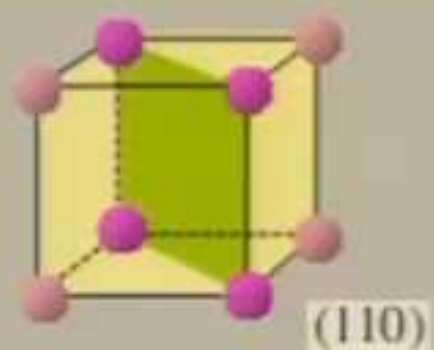
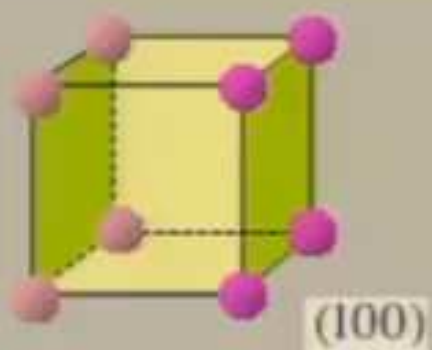


FCC = Simple Cubic with Basis

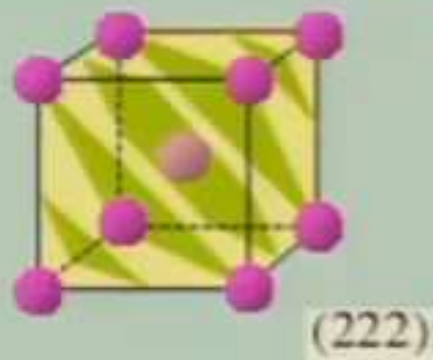
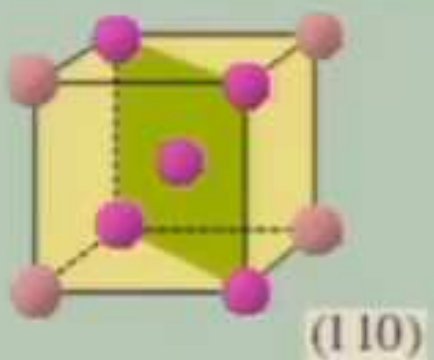
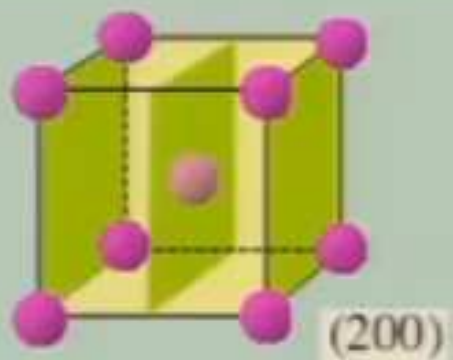
Basis =  $[0,0,0]$ ,  $[\frac{1}{2}, \frac{1}{2}, 0]$ ,  $[\frac{1}{2}, 0, \frac{1}{2}]$ ,  $[0, \frac{1}{2}, \frac{1}{2}]$



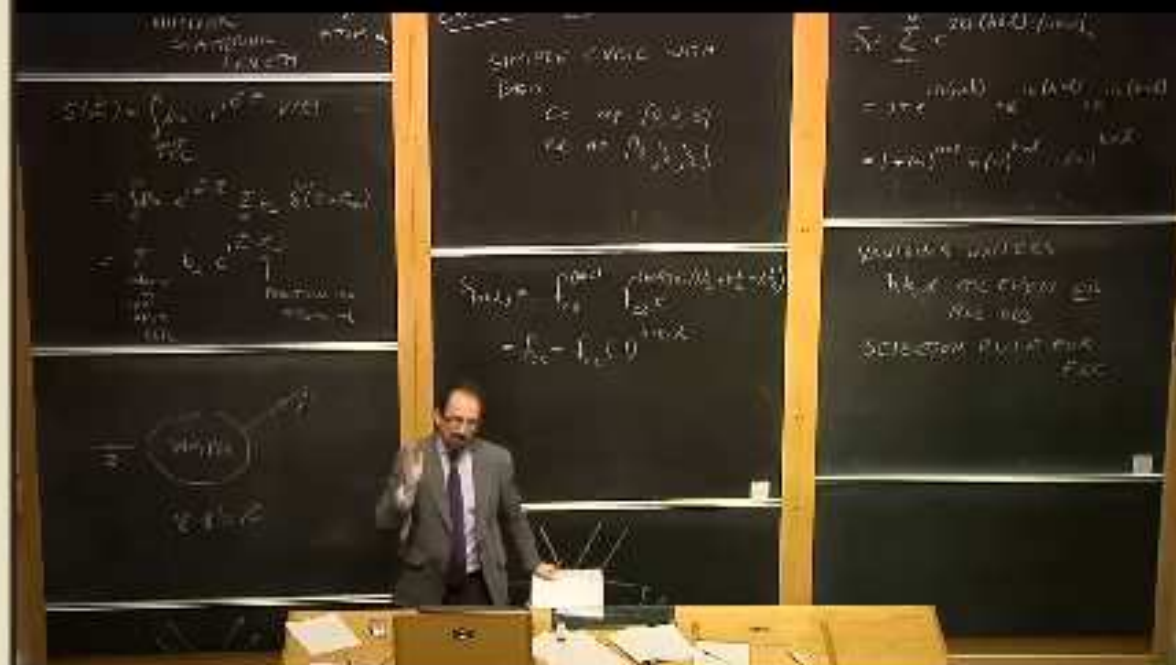
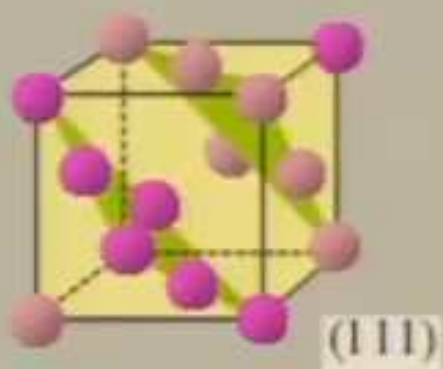
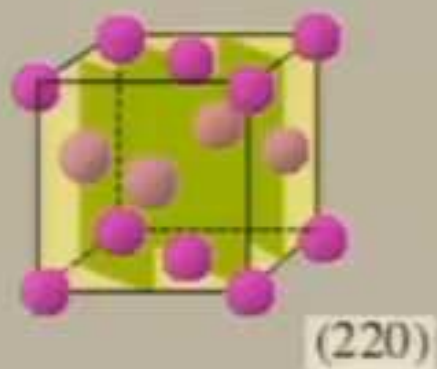
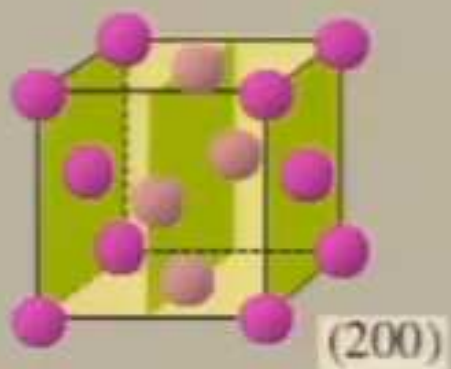
SC



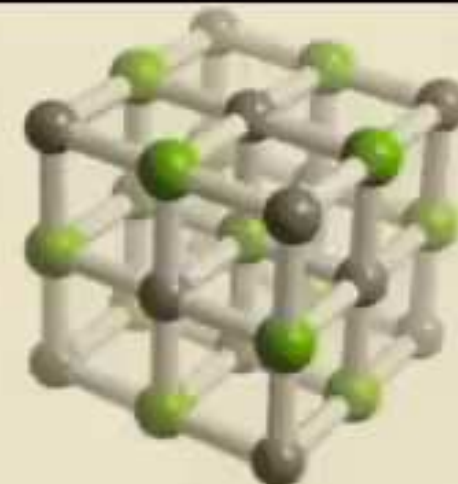
bcc



fcc







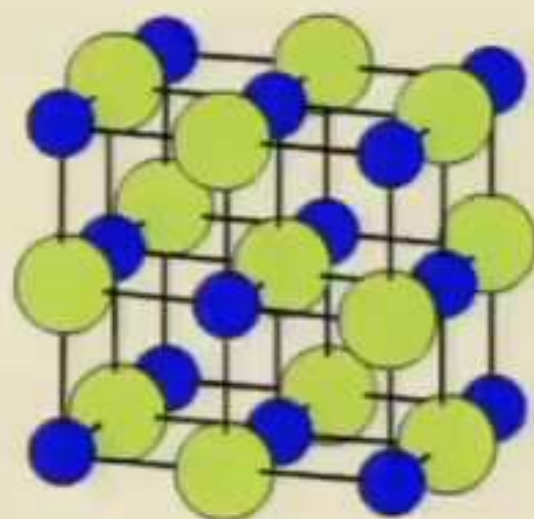
## sodium chloride (NaCl)

lattice: cubic F

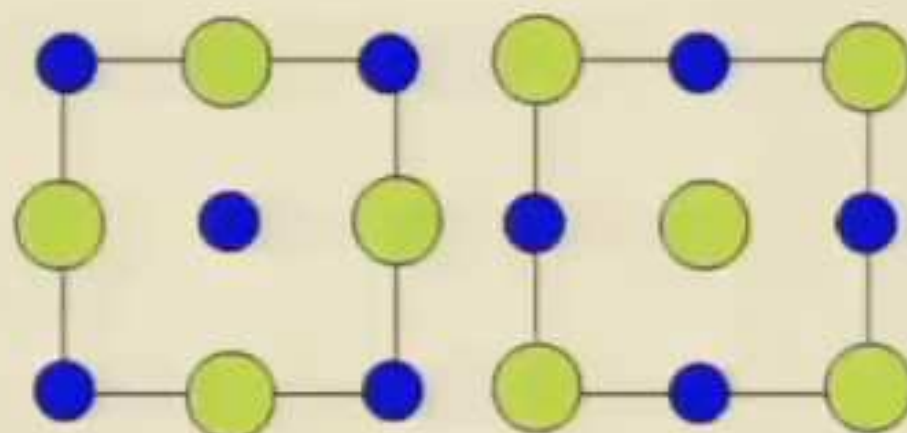
basis :

Na 000

Cl  $\frac{1}{2}\frac{1}{2}\frac{1}{2}$



Plan view

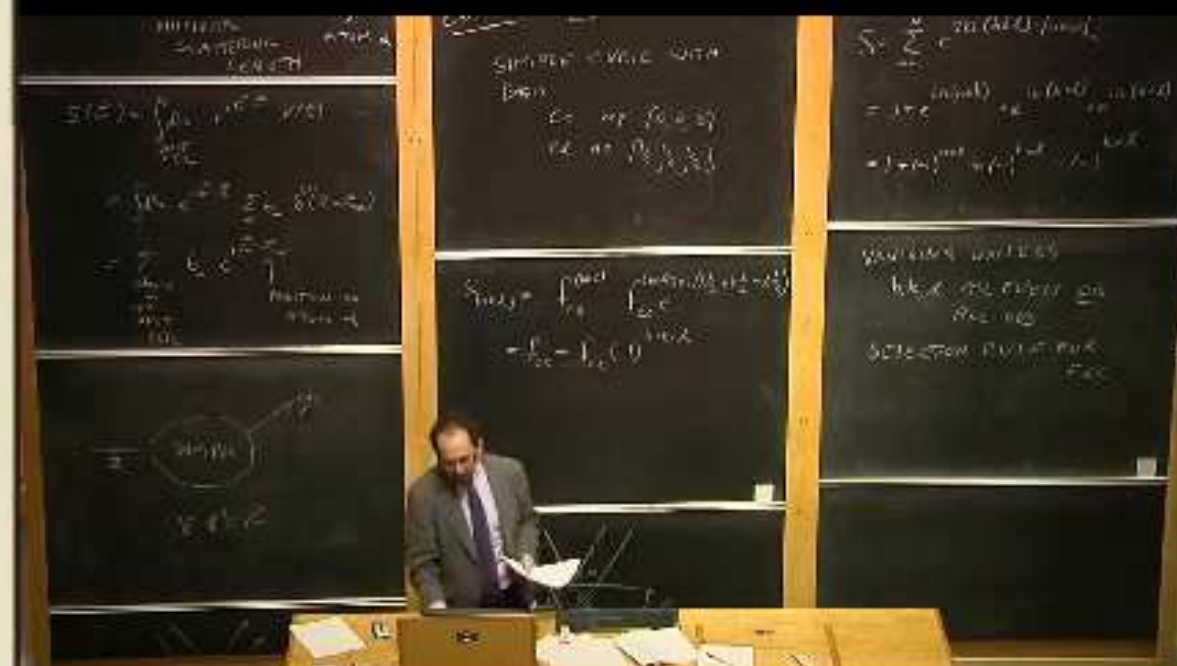


$z = 0$  layer

$z = \frac{1}{2}$  layer

Na forms FCC lattice.

Cl is displaced  $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$  from each Na



Pictures for Lecture 12b.pptx - Microsoft PowerPoint

Home Insert Design Transitions Animations Slide Show Review View

Slide 1: Condensed Matter Physics Lecture 12

Slide 2: [Diagram of a crystal lattice]

Slide 3: [Diagram of a crystal lattice]

Slide 4: [Diagram of a crystal lattice]

Slide 5: [Diagram of a crystal lattice]

Slide 6: [Diagram of a crystal lattice]

Slide 7: [Diagram of a crystal lattice]

Slide 8: [Diagram of a crystal lattice]

Slide 9: [Diagram of a crystal lattice]

Slide 10: [Diagram of a crystal lattice]

Slide 11: [Diagram of a crystal lattice]

Slide 12: [Diagram of a crystal lattice]

Slide 13: [Diagram of a crystal lattice]

Slide 14: [Diagram of a crystal lattice]

Slide 15: [Diagram of a crystal lattice]

Slide 16: [Diagram of a crystal lattice]

Slide 17: [Diagram of a crystal lattice]

Slide 18: [Diagram of a crystal lattice]

Slide 19: [Diagram of a crystal lattice]

Slide 20: [Diagram of a crystal lattice]

Handwritten notes on blackboards:

Slide 1: [Diagram of a crystal lattice]

Slide 2: [Diagram of a crystal lattice]

Slide 3: [Diagram of a crystal lattice]

Slide 4: [Diagram of a crystal lattice]

Slide 5: [Diagram of a crystal lattice]

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Slide 17: [Diagram of a crystal lattice]

Slide 18: [Diagram of a crystal lattice]

Slide 19: [Diagram of a crystal lattice]

Slide 20: [Diagram of a crystal lattice]



Pictures for Lecture 12b.pptx - Microsoft PowerPoint

Home Insert Design Transitions Animations Slide Show Review View

6 7 8 9 10

11 12 13 14 15

16 17 18 19 20

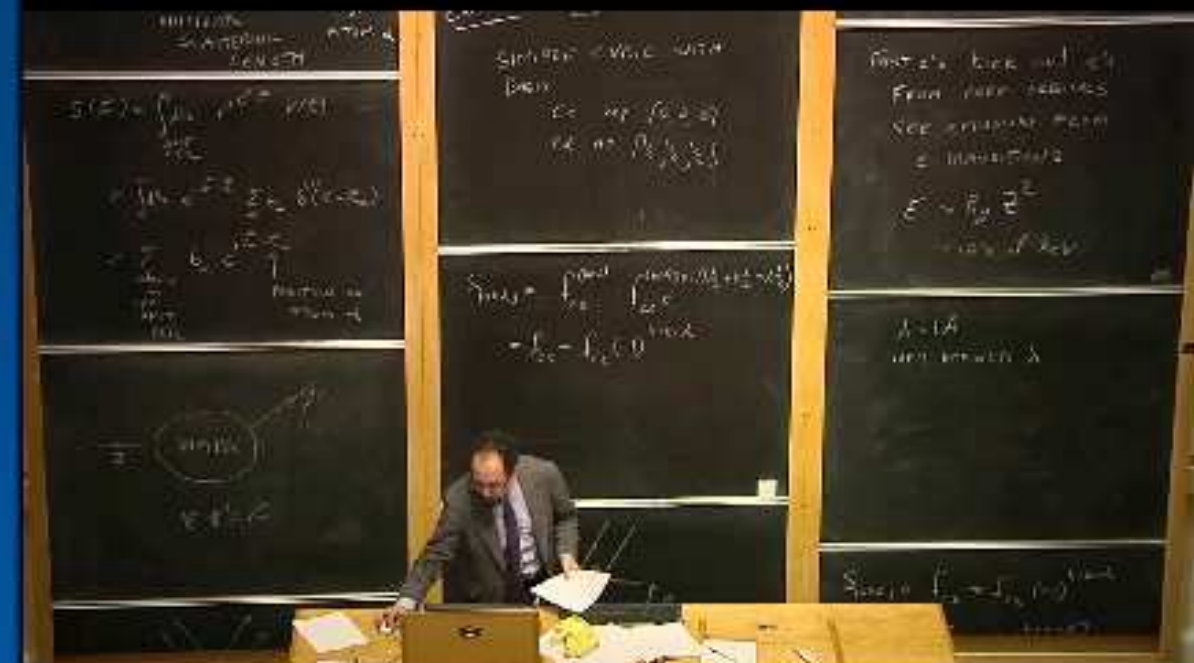
21 22 23 24 25

26 27 28 29 30

Office Theme

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# The Rutherford-Appleton Lab in Oxfordshire



Spallation Neutron Source

Synchrotron X-ray Source

